

Md Mobarak Karim

Postdoctoral Associate | Biomedical Optics & Computational Microscopy

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Academic Profile

Postdoctoral Associate in the Richards-Kortum Laboratory at Rice University with a Ph.D. in Biomedical Engineering and more than six years of experience in biomedical optics and computational microscopy. Current research focuses on translational microscopy for digital pathology, including AccessPath and PrecisionView, Zemax optical design, automated XYZ microscopy, array microscopy, phase-mask optics, virtual H&E imaging, autofocus, image stitching, and wavelength/contrast optimization. Doctoral research established co-registered OCT-LSFM systems for longitudinal structural and molecular imaging of mouse and zebrafish embryonic development, with contributions spanning optical engineering, quantitative tissue optics, image analysis, biological validation, mentoring, and scientific dissemination.

Research Expertise

Biomedical Imaging	Swept-source OCT, spectral-domain OCT, OCT angiography, optical coherence elastography, one- and two-photon LSFM, multiphoton microscopy, fluorescence microscopy, computational microscopy, array microscopy
Optical System Development	Zemax OpticStudio, optical alignment, Gaussian beam propagation, beam expanders, 4f relays, scan/tube lenses, galvo scanning, fiber optics, polarization control, phase-mask optics, laser-camera synchronization
Computational Pathology	Virtual H&E imaging, wavelength and contrast optimization, background correction, frame averaging, image stitching, autofocus, automated XYZ acquisition, image-quality assessment
Quantitative Imaging	Attenuation-coefficient mapping, sensitivity roll-off correction, confocal correction, morphometry, segmentation, multimodal co-registration, 3D/4D reconstruction, statistical analysis
Programming and Software	MATLAB, Python, C/C++, LabVIEW, ImageJ/Fiji, Imaris, Amira, Optiland, GUI development, automated acquisition and processing workflows
Biological Applications	Digital pathology, mouse and zebrafish embryonic development, vascular imaging, prenatal alcohol exposure, developmental toxicity, live-sample imaging, optical clearing

Academic and Research Appointments

Postdoctoral Associate - Biomedical Optics and Computational Microscopy	Aug 2026–Present
<i>Richards-Kortum Laboratory, Department of Bioengineering, Rice University</i>	<i>Houston, TX</i>
<i>Principal Investigator: Dr. Rebecca Richards-Kortum</i>	

- Develop optical and computational imaging technologies for **AccessPath**, with emphasis on translational microscopy and virtual histology for pathology applications.
- Design and evaluate AccessPath optics in **Zemax OpticStudio**, including lens and relay configurations, component selection, alignment tolerances, image-quality assessment, and experimental validation.
- Developed an integrated unit-microscopy platform combining motorized **X/Y/Z acquisition**, autofocus, frame averaging, background correction, image stitching, and parallel virtual H&E processing within a unified Python GUI.
- Lead the **AccessPath APHA-H Phase II optics team**, coordinating optical architecture, phase-mask design and fabrication, component evaluation, and validation for **array microscopy** development.
- Contribute to the **PrecisionView** optics team through optical-system integration, characterization, troubleshooting, and imaging-performance optimization.
- Investigate **novel contrast mechanisms and excitation/emission wavelength combinations** for virtual H&E imaging to improve tissue contrast, reduce optical background, and enhance pathology-relevant image quality.
- Collaborate with multidisciplinary teams in optics, instrumentation, software, image processing, and pathology to translate prototype systems into reproducible imaging workflows.

Graduate Research Assistant - Multimodal Optical Imaging	Jan 2021–May 2026
<i>Biomedical Optics Laboratory, University of Houston</i>	<i>Houston, TX</i>
<i>Advisor: Dr. Kirill V. Larin</i>	

- Designed and integrated a co-registered multimodal platform combining **1051-nm swept-source OCT** with **one-photon and two-photon LSFM** for simultaneous label-free structural and fluorescence imaging of mouse and zebrafish embryos.
- Engineered beam delivery, polarization control, galvo scanning, scan-lens interfaces, acquisition timing, camera/laser synchronization, sample positioning, and environmental control for stable 3D and longitudinal imaging.
- Characterized system performance, achieving approximately **7–7.6 μm OCT axial resolution**, **15 μm lateral resolution**, approximately **100 dB sensitivity**, and approximately **6.9 dB roll-off over 4.13 mm**.

- Demonstrated approximately **250 μm** one-photon fluorescence penetration and greater than **800 μm** two-photon visualization in scattering embryonic tissue while OCT provided whole-embryo morphology.
- Developed depth-resolved OCT attenuation algorithms incorporating noise subtraction, confocal correction, and sensitivity-roll-off compensation; validated the method using multilayer TiO_2 phantoms and applied it to murine embryonic brain development.
- Built MATLAB and Python workflows for OCT reconstruction, multimodal co-registration, attenuation mapping, morphometry, segmentation, 3D/4D visualization, image-quality analysis, and standardized processing.
- Established long-term live-imaging workflows and a low-cost environmental incubator for OCT-LSFM studies; applied the platform to zebrafish development, vascular imaging, and prenatal ethanol-exposure experiments.
- Led optical integration, experimental execution, troubleshooting, data analysis, and manuscript preparation; mentored **three Ph.D. students and several undergraduate researchers** in optical imaging and quantitative analysis.

Graduate Research Assistant - Optical Design and Two-Photon Microscopy

3D Biomedical Imaging and Technology Laboratory, GIST

Aug 2018–Dec 2020

Gwangju, South Korea

- Designed and optimized broadband scan and tube lenses for laser-scanning microscopy using **Zemax OpticStudio**; evaluated wavefront error, Strehl ratio, aberrations, telecentricity, field of view, and F-theta distortion across 680–1600 nm.
- Developed four optimized scan lenses and one tube lens, reducing F-theta distortion from 13.00 to 1.94 and telecentric distortion from 3.52 to 0.4367.
- Supported development and qualification of a two-photon laser-scanning microscope through optical component evaluation, alignment, testing, measured-performance analysis, and structured troubleshooting.

Education

Ph.D., Biomedical Engineering

University of Houston

Jan 2021–May 2026

Houston, TX

Advisor: Dr. Kirill V. Larin. Dissertation: *OCT-LS: A multimodal co-registered optical coherence tomography and light-sheet fluorescence microscopy platform for simultaneous structural and molecular imaging of embryonic development.*

M.S., Biomedical Science and Engineering

Gwangju Institute of Science and Technology

Aug 2018–Dec 2020

Gwangju, South Korea

Advisor: Dr. Hyuk-Sang Kwon. Thesis: *Analysis of an achromatic doublet as a scan lens for laser-scanning microscopy.*

B.S., Electrical and Electronic Engineering

International Islamic University Chittagong

Aug 2013–Aug 2017

Chittagong, Bangladesh

Thesis: *Feasibility analysis and proposal for a 1.3-MW hybrid renewable power plant for Saint Martin's Island, Bangladesh.*

Peer-Reviewed Journal Articles

- **Md Mobarak Karim**, Ruijiao Sun, Behzad Khajavi, Manmohan Singh, Yogeshwari S. Ambekar, Alexander W. Schill, Salavat R. Aglyamov, David Mayerich, and Kirill V. Larin. "Multimodal optical coherence tomography and two-photon light-sheet fluorescence microscopy for embryo imaging." *Journal of Biomedical Optics* 30(6), 060501 (2025). DOI: 10.1117/1.JBO.30.6.060501.
- **Md Mobarak Karim**, Achuth Nair, Manmohan Singh, Maryam Hatami, Salavat R. Aglyamov, and Kirill V. Larin. "Depth-resolved attenuation coefficient quantification during murine embryonic brain development." *Journal of Biophotonics* 18(10), e202500212 (2025). DOI: 10.1002/jbio.202500212.
- Leah A. Lewis, **Md Mobarak Karim**, et al. "Multimodal optical imaging for the assessment of the teratogenic effects of ethanol on zebrafish development." *Journal of Biomedical Optics* 31(6), 066004 (2026). DOI: 10.1117/1.JBO.31.6.066004.
- Mumtaza Mumu, Ayan Das, Talha Bin Emran, Saikat Mitra, Fahadul Islam, Arpita Roy, **Md Mobarak Karim**, Rajib Das, Moon Nyeo Park, Deepak Chandran, Rohit Sharma, Mayeen Uddin Khandaker, Abubakr M. Idris, and Bonglee Kim. "Fucoxanthin: A promising phytochemical on diverse pharmacological targets." *Frontiers in Pharmacology* 13, 929442 (2022). DOI: 10.3389/fphar.2022.929442.

Manuscripts in Preparation or Under Review

- **Md Mobarak Karim**, et al. "Long-term imaging of zebrafish embryonic development with multimodal optical coherence tomography and light-sheet fluorescence microscopy." Manuscript in preparation.
- James L. Klein, **Md Mobarak Karim**, Christian Zevallos-Delgado, Manmohan Singh, Salavat R. Aglyamov, and Kirill V. Larin. "Tartrazine-based optical clearing for monitoring embryogenesis with optical coherence tomography." Submitted to *Optics Letters*.

Selected Conference Proceedings

- **Md Mobarak Karim**, Leah A. Lewis, Geethangili Madamanchi, Manmohan Singh, Alexander W. Schill, Salavat R. Aglyamov, David Mayerich, Arne C. Lekven, and Kirill V. Larin. "OCT-LS: A co-registered OCT and LSFM platform for simultaneous structural and molecular imaging of embryonic development." *Proceedings of SPIE* 13841, Multimodal Biomedical Imaging XXI (2026).

DOI: 10.1117/12.3080842.

- **Md Mobarak Karim**, Leah A. Lewis, Christian Zevallos-Delgado, Oscar E. Ruiz, Geethangili Madamanchi, Alexander W. Schill, Salavat R. Aglyamov, Arne C. Lekven, and Kirill V. Larin. “Combined structural and molecular zebrafish embryonic imaging on ethanol exposure effects during vasculature development.” *Proceedings of SPIE* 13327, Multiscale Imaging and Spectroscopy VI (2025). DOI: 10.1117/12.3043312.
- **Md Mobarak Karim**, Ruijiao Sun, Oscar E. Ruiz, Leah A. Lewis, Manmohan Singh, Alexander W. Schill, Salavat R. Aglyamov, David Mayerich, and Kirill V. Larin. “Imaging zebrafish embryonic development with multimodal optical coherence tomography and light-sheet fluorescence microscopy.” *Proceedings of SPIE* 12834, Multimodal Biomedical Imaging XIX (2024). DOI: 10.1117/12.3001234.
- **Md Mobarak Karim**, Ruijiao Sun, Behzad Khajavi, Manmohan Singh, Harshdeep S. Chawla, Yogeshwari S. Ambekar, Alexander W. Schill, David Mayerich, Mary E. Dickinson, and Kirill V. Larin. “Multimodal optical coherence tomography and two-photon selective-plane illumination microscopy for embryonic imaging.” *Proceedings of SPIE* 12371, Multimodal Biomedical Imaging XVIII (2023). DOI: 10.1117/12.2649508.
- **Md Mobarak Karim**, Ruijiao Sun, Behzad Khajavi, Manmohan Singh, Harshdeep S. Chawla, David Mayerich, Mary E. Dickinson, and Kirill V. Larin. “Combined optical coherence tomography and light-sheet fluorescence microscopy for embryonic imaging.” *Proceedings of SPIE* 11952, Multimodal Biomedical Imaging XVII (2022). DOI: 10.1117/12.2610091.

Advanced Research Training

- **Multiscale Biophotonics Training Program**, Beckman Laser Institute, University of California, Irvine, 2024. NIH-supported training in fluorescence lifetime imaging, nonlinear microscopy, spectroscopy, and translational biophotonics.
- **CLIMB Biophotonics Summer School**, Beckman Institute, University of Illinois Urbana-Champaign, 2023. Hands-on OCT/OCM system construction, reconstruction, and quantitative imaging.
- **Siegman International School on Lasers**, Poland, 2022. Advanced training in laser physics, ultrafast optics, nonlinear optics, and photonic systems.

Selected Honors and Awards

2025	Cover image featured in <i>Journal of Biomedical Optics</i>
2024	NIH Travel Grant, UCI Multiscale Biophotonics Training Program
2022	Zuegel Family Scholarship, Siegman International School on Lasers
2022	OPTICA Siegman International School on Lasers Grant
2022	SPIE Photonics West Travel Grant
2020	Outstanding Poster Award, Biomedical Research Meeting, South Korea
2018–2020	Korean Government Scholarship for M.S. study at GIST
2013–2017	Undergraduate Merit Award and Scholarship, Department of EEE, IIUC

Mentoring and Teaching

- Mentored three Ph.D. students and several undergraduate researchers in OCT/LSFM alignment, microscopy operation, image acquisition, MATLAB/Python analysis, experimental design, and scientific presentation.
- Trained junior researchers in system calibration, image-quality evaluation, multimodal co-registration, segmentation, and 3D visualization.
- Supported multidisciplinary collaboration among biomedical engineers, optical engineers, developmental biologists, and imaging scientists.

Professional Service and Leadership

- Reviewer for biomedical optics and photonics journals.
- Senior Judge, Texas Science and Engineering Fair - Biomedical Division (2022).
- Executive Member, University of Houston SPIE Student Chapter (2021–2026).
- Member, OPTICA (formerly the Optical Society of America) (2021–Present).
- Founder and former executive member, Bangladesh Robotics Foundation.
- Assistant General Secretary, IIUC EEE Club (2017); Troop Leader, Bangladesh Red Crescent Youth; Leadership Graduate, BYLC Art of Leadership Program (2016).

References

Available upon request. A separate reference list can be provided for academic applications.